

Bilal Tariq

(413) 472-4746 | mtariq27@amherst.edu | linkedin.com/in/tariq-bilal | github.com/bilaltiq | bilaltariq.net

Education

Amherst College

B.A. in Computer Science & Mathematics (Double Major)

GPA: 3.86/4.0 · Major GPA: 3.91/4.0

Expected May 2027

Amherst, MA

- **Coursework:** Data Structures, Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, Natural Language Processing, Real Analysis, Linear Algebra
- **Honors & Leadership:** Meiklejohn Fellowship (\$5,000 startup R&D grant); AI in the Liberal Arts Fellow; E-Board, CS Club; Core Organizer, “Hack The Herd”

Experience

Finosu (YC S24)

Software Engineering Intern

Python, TypeScript, LLMs, PostgreSQL, AWS S3, React, Git

June 2026 – Aug 2026

New York, NY

- Built the unified conversation-memory system behind **1.4M+** calls, now live for every borrower: compresses multi-call history into prompt-sized rolling summaries with live-state reconciliation, shipped through a shadow-mode A/B that raised mean call-quality score **70%** over the prior pipeline
- Shipped tool-calling AI chatbots across 12 lenders’ borrower portals serving **151.6K** customers, owning the agent tool layer — persistent knowledge memory with S3 document and live database tool calls — so replies resolve against the borrower’s real account state rather than static prompt text
- Shipped two production LLM services: a login-triggered outbound-call service whose A/B rollout measured 4× on-time loan repayment against control, and a chained-LLM triage pipeline that auto-escalates post-call issues and removed **~40** analyst hours/week of manual QA
- Consolidated inbound and outbound voice agents onto one configuration library, making LLM/STT/TTS swaps and live model A/B comparison possible without a deploy, and closed a **15-CVE** sweep (XSS, prototype pollution, DoS) that hardened unauthenticated admin endpoints over borrower data

Layout

Co-Founder & CTO

TypeScript, AWS Lambda, DynamoDB, Cognito, API Gateway, React/Next.js

Oct 2024 – Present

Amherst, MA

- Architected a production 3-D interior-design SaaS (withlayout.com) on serverless AWS — API Gateway with a custom Lambda authorizer, DynamoDB schemas, and S3 storage provisioned as code via AWS CDK, secured with OAuth 2.0 + PKCE on Cognito and server-side JWT validation
- Recruited and led a **7-engineer** team over **20** months on push-to-main CI/CD, delivering full-stack SaaS spanning auth, Stripe billing, AI furniture generation, and Babylon.js 3-D rendering

Google exploreCSR — Brown University

Undergraduate AI Researcher

C++/CUDA, Python, PyTorch Geometric, Git

Jan 2025 – Apr 2025

Providence, RI

- Rebuilt a CAD-to-sketch neural rendering pipeline around Graph Attention Networks with custom CUDA/C++ ops, leading implementation for a 5-person research group; profiling and multi-head-attention optimizations cut convergence time **17%** and raised accuracy from 87% to **93%**
- Won **1st place** at the Brown & Google Undergraduate CS Research Symposium (40 researchers, \$1,000 award)

Projects

Probing Semantic Interpretability in Multilingual LLMs | *PyTorch, HuggingFace Transformers, scikit-learn*

2025

- Trained **6,500+** layer-wise linear probes across BERT, mBERT, and XLM-R over 50 languages, locating where 5 AMR semantic roles become linearly recoverable across 13 encoder layers of the 84K-annotation MASSIVE-AMR dataset
- Recovered agent (ARG0) roles at up to **0.90** accuracy / 0.86 F1 and showed multilingual encoders hold semantics for lower-resource languages roughly 5 layers deeper than for English

SiFT — Secure File Transfer Protocol | *Python, AES-GCM, RSA-OAEP, HKDF, TCP Sockets*

2025

- Built a secure client-server file-transfer system implementing 5 sub-protocols and 7 remote file commands over TCP with AES-256-GCM authenticated encryption, RSA-2048-OAEP key exchange, and HKDF-SHA256 session-key derivation
- Hardened the message layer against eavesdropping, tampering, and replay via explicit sequence numbers, per-message nonces, and MAC verification on every frame

Technical Skills

Languages: C++, C, Python, Java, TypeScript/JavaScript, SQL, RISC-V Assembly

Systems & Infrastructure: Linux, Concurrency, Memory Management, CUDA, Networking, Cryptography, Docker, AWS (Lambda, DynamoDB, S3, API Gateway, Cognito), PostgreSQL, Git, GitHub Actions CI/CD

ML & Tooling: PyTorch, TensorFlow, Keras, PyTorch Geometric, HuggingFace Transformers, scikit-learn, LLM tool-calling & retrieval; Claude Code, Cursor, GitHub Copilot